

PART IV

Money and Inflation

This part of the book looks at the role of money in the economy.

In Chapter 10 we study what money is, how the quantity of money is determined and why people choose to hold it.

In Chapter 11 we study how the equilibrium in the money market is determined, what determines the price level and how inflation comes about.

CHAPTER 10

Money

10.1 What is Money?

This question is trickier than it seems.¹ The standard answer is that “money” can be anything that can serve as:

1. A store of value. One can save it (for instance, by keeping it in one’s pocket) and use it later.
2. A unit of account. We can express the prices of things in terms of how much money it takes to buy them.
3. A medium of exchange. Money changes hands when people pay for things.

Many different things have been used as money at various times and places: pieces of paper with the faces of historical figures, gold and silver, cigarettes.

Why do we use money at all? The main reason is that it solves what’s known as the “double coincidence of wants” problem. Using money, I don’t need to find someone who has exactly what I need and wants exactly what I have in order to trade. I can accept money in payment for the goods I sell knowing that others will accept money in payment for the goods I want.

For something to be convenient to use as money, it typically needs to have several properties:

1. It has to be hard to counterfeit. If you want to pay for something with money, you don’t want the seller to be wondering whether you are giving them real money or fake money.
2. It has to be easy to carry, since transactions happen in many different places.
3. It has to be durable, otherwise it’s not a very good store of value. It makes more sense to use coffee beans than strawberries as money because strawberries are likely to spoil before they can be used in the next transaction.
4. It has to be easily divisible. For transactions to go smoothly it’s important to be able to pay the exact price without too much rounding and to be able to make change. The wonderful book by Sargent and

¹ Asmundson and Oner (2012) have a good introduction to this question.

Velde (2014): *The big problem of small change* tells the history of how Europe dealt with the problem of making change.

- 5. It has to be commonly accepted. Money is only useful if everyone agrees that it is indeed money and accepts it as payment. Sometimes this acceptance is purely a social convention, sometimes it is reinforced by laws.

The properties listed above are satisfied to different degrees by different assets. As a result, there is no unique measure of what is the type of money that is used in any economy or how much of it there is. By convention, several definitions of money are typically studied, each of which draws a somewhat arbitrary line between “money” and “not money”. Here are the main ones:

Monetary base	M0	M1	M2
Physical Currency	Physical Currency	Physical Currency	Physical Currency
Central Bank Reserves		Demand deposits	Demand deposits
			Savings deposits
			Small time deposits
			Money market mutual funds

Let’s start from M0. This measure counts as money only physical bills and coins. It is quite clear that physical currency meets the conditions for something to be money pretty well. Not perfectly, though: it’s possible to counterfeit and there are some places where it’s not accepted as a means of payment. M1 is a broader measure of money because it also includes demand deposits (basically checking accounts). For most purposes, a checking account satisfies the definition of money: for most transactions, either a check or a debit card is acceptable.² M2 is a still broader measure because it includes other types of deposits (savings accounts and smaller time deposits held by individuals) as well as shares in money market mutual funds held by individuals.³ The assets in this broader measure of money are slightly less easy to use in transactions. That’s why these assets have an intermediate degree of “moneyness” and are included in the broader measures of money but not the narrower ones.

The “monetary base” has this name because it’s the only part of money that is under the direct control of the government. Therefore it forms the “base” on which every other measure is built. To see this, it is useful to go into the mechanics of how the quantity of money is determined.

10.2 The Supply of Money

The monetary aggregates defined above are in part chosen by the government, which decides how much physical currency to issue, and in part determined by what happens in the banking sector, since deposits are bank liabilities. How does this all fit together?

²But not for every transaction. Some stands in the Palo Alto farmers’ market will only take physical cash. So (I’ve been told) will most drug dealers.

³A mutual fund is an investment vehicle by which investors each own a proportion of a pool of assets. A mutual fund is called a “money market” fund if it invests in very safe assets so that the total value of the pool of assets doesn’t move much, making it very similar to money as a store of value.

To address this, a good place to start is by looking at what a bank balance sheet looks like. Here is a typical bank balance sheet:

Balance Sheet	
Assets	Liabilities
Reserves	Deposits
Bonds	
Loans	Other Liabilities
Other Assets	Net Worth

On the left are all the bank's assets: loans, government bonds, etc. One of the main sources of income for banks is the interest it earns on these assets (the other is fees of various kinds). One asset in particular will be relevant to us: Central Bank reserves. The Central Bank acts as a bank for banks and reserves is just the name given to the deposits that banks hold at the Central Bank (i.e. this is an asset for banks and a liability for the Central Bank). Typically, these reserves earn either no interest or a very low rate.⁴ Why do banks hold them?

There are two reasons, whose relative importance has been different at different times in history. One reason is that reserves are a way to meet unexpected withdrawals of deposits. Most bank assets are relatively long term and hard to sell so if depositors want their money right away it's useful for the bank to have an asset that can be converted into cash very quickly, and reserves provide this: in most countries, the Central Bank stands ready to exchange reserves for cash whenever banks want it. Nowadays this is usually not the main reason banks keep reserves since deposit insurance has made bank runs quite rare and there are explicit arrangements for banks to get emergency loans to meet deposit withdrawals. Instead, the main reason banks keep reserves is that they are required to do so by regulation. Typically, banks are required to hold a certain minimum level of reserves, set as a percentage of the bank's deposits. The exact percentage usually depends on the type of deposit, though the details vary a lot from one country to another.

On the right hand side are the bank's liabilities: mostly deposits but sometimes also non-deposit borrowing such as long term bonds that the bank has issued. The bank's net worth is the difference between its assets and liabilities.

Let's compute the supply of money in a simplified example. There are three relevant entities:

- the Central Bank,
- a single commercial bank, which is meant to represent the sum of all banks in the economy,
- a single representative household.

These are their balance sheets:

⁴This is changing. In recent years many central banks are paying interest on reserves. We look into this further below.

Central Bank		Private Bank		Household	
Assets	Liabilities	Assets	Liabilities	Assets	Liabilities
Bonds: b	Reserves: ρd	Reserves: ρd	Deposits: d	Currency: c	0
	Currency: c	Bonds: B		Deposits: d	
	Net Worth	Loans: L	Net Worth		Net Worth
	$b - \rho d - c$		$\rho d + B + L - d$		$c + d$

The household owns currency and deposits and, in this example, has no liabilities or any other assets. The bank’s assets are made up of loans, government bonds and Central Bank reserves. The reserve requirement in this example is set at a fraction ρ of deposits and the bank is satisfying it exactly, so it has ρd in reserves. The Central Bank owns a certain amount of government bonds b and its liabilities are the private bank’s reserves and the outstanding currency.⁵

In this economy we have that, following the definitions above:

- the monetary base is $c + \rho d$,
- M0 is c ,
- M1 is $c + d$ (assuming the deposit is a demand deposit).

10.3 Changing the Supply of Money

In this section we’ll study the traditional way in which central banks would change the supply of money. The way central banks operate has been changing over the last decade or so. Exercise 10.3 asks you to think about how more modern operating procedures compare with this traditional approach.

Suppose that the Central Bank wants to change the supply of M1 money. We’ll come to the reasons why the Central Bank might want to do that later on, but for now let’s just accept that the Central Bank wants to do this. The Central Bank doesn’t directly control the amount of deposits, which represent the biggest component of M1, but it affects it indirectly through “open market operations”. Let’s see how this works.

An open market operation is a trade by the Central Bank where the Central Bank either buys bonds and pays for them with reserves or sells bonds and accepts reserves as payment. It’s called open market operation because the Central Bank is trading just like anyone else in the open market. Let’s work through how an open market operation takes place in the example above. Suppose that the Central Bank buys government bonds worth Δ from the private bank and pays for them by crediting Δ reserves to the private bank. These are new reserves: they come into existence because the Central Bank creates them. Balance sheets are now:

⁵Thinking of currency as a liability of the Central Bank is a bit counterintuitive at first: the Central Bank doesn’t really have an obligation to pay anything to holders of currency. This wasn’t always the case: it used to be that currency represented a promise by the Central Bank to deliver gold to the holder. Then currency was in every sense a liability, and the accounting reflected that.

Central Bank		Private Bank		Household	
Assets	Liabilities	Assets	Liabilities	Assets	Liabilities
	Reserves:	Reserves:	Deposits: d	Currency: c	0
	$\rho d + \Delta$	$\rho d + \Delta$			
Bonds: $b + \Delta$	Currency: c	Bonds: $B - \Delta$			
	Net Worth	Loans: L	Net Worth		Net Worth
	$b - \rho d - c$		$\rho d + B + L - d$		$c + d$

But this is not the end of the story. Now the private bank has reserves of $\rho d + \Delta$ but it's only required to have ρd . It now has "excess reserves". Since reserves don't earn any interest, the bank will try to lend out these excess reserves.⁶ Let's imagine that the bank makes a loan of Δ to the household. What exactly happens when the bank makes a loan?

- The bank gives the borrower a check for Δ in exchange for a promise that the borrower will pay it back with interest later.
- The borrower deposits the check. Maybe the borrower deposits it in the same bank; maybe the borrower deposits it in a different bank; maybe the borrower hands over the check to someone else (for instance, someone who sells him a car) and then that person deposits the check in their bank. Since we are adding up over all banks and over all households, all these variants are equivalent.

The balance sheets are now:

Central Bank		Private Bank		Household	
Assets	Liabilities	Assets	Liabilities	Assets	Liabilities
	Reserves:	Reserves:	Deposits:	Currency: c	Loans: Δ
	$\rho d + \Delta$	$\rho d + \Delta$	$d + \Delta$		
Bonds: $b + \Delta$	Currency: c	Bonds: $B - \Delta$		Deposits:	
		Loans: $L + \Delta$		$d + \Delta$	
	Net Worth	Net Worth			Net Worth
	$b - \rho d - c$	$\rho d + B + L - d$			$c + d$

Notice that even though the bank "lent out the excess reserves" the reserves don't actually disappear. They are still there. It's more accurate to say: the private banks take advantage of the relaxation of the reserve-to-deposit ratio to expand loans and deposits.

Now suppose that the borrower (or anyone that the borrower made a payment to) wants to take out a fraction of the Δ new deposits in cash. Call that fraction χ , so that the borrower wants to have $\chi\Delta$ cash and $(1 - \chi)\Delta$ deposits. The borrower goes to the ATM and makes a withdrawal. What exactly happens when the borrower does this?

- The bank asks the Central Bank for cash (note that in our example the bank held zero cash to begin with).

⁶This assumes that there are lending opportunities out there where the bank will in fact earn a positive interest rate. Later we'll think about what happens if interest rates are zero, or when the Central Bank pays interest on reserves.

10.3. Changing the Supply of Money

- The Central Bank prints physical currency and gives it to the bank. In return, it reduces the amount of reserves owed to the bank.
- The bank hands over the cash to the borrower. In return, it reduces the balance on the borrower's deposit.

The balance sheets are now:

Central Bank		Private Bank		Household	
Assets	Liabilities	Assets	Liabilities	Assets	Liabilities
	Reserves:	Reserves:	Deposits:	Currency:	Loans: Δ
	$\rho d + (1 - \chi) \Delta$	$\rho d + (1 - \chi) \Delta$		$c + \chi \Delta$	
Bonds: $b + \Delta$	Currency:	Bonds: $B - \Delta$	$d + (1 - \chi) \Delta$	Deposits:	
	$c + \chi \Delta$	Loans: $L + \Delta$			
	Net Worth		Net Worth	$d + (1 - \chi) \Delta$	Net Worth
	$b - \rho d - c$		$\rho d + B + L - d$		$c + d$

Notice that nobody's net worth changes in this whole series of transactions. The participants are just exchanging different types of assets and liabilities with each other.

The process is not over. The bank has $\rho d + (1 - \chi) \Delta$ reserves but the reserve requirement is only $\rho(d + (1 - \chi) \Delta)$ so there are excess reserves of $(1 - \rho)(1 - \chi) \Delta$. Therefore the process repeats itself, just scaled down by $(1 - \rho)(1 - \chi)$, and then by $((1 - \rho)(1 - \chi))^2$, and then by $((1 - \rho)(1 - \chi))^3$, etc. This defines a geometric series, so we can compute the total effect as follows:

$$\text{Change in Deposits} = \sum_{n=0}^{\infty} ((1 - \rho)(1 - \chi))^n \underbrace{(1 - \chi) \Delta}_{\text{First round effect}} = \frac{(1 - \chi) \Delta}{\rho + \chi - \rho\chi}$$

$$\text{Change in Currency Holdings} = \sum_{n=0}^{\infty} ((1 - \rho)(1 - \chi))^n \underbrace{\chi \Delta}_{\text{First round effect}} = \frac{\chi \Delta}{\rho + \chi - \rho\chi}$$

$$\text{Change in Reserves} = \Delta - \text{Change in Currency Holdings} = \frac{\rho(1 - \chi) \Delta}{\rho + \chi - \rho\chi}$$

$$\text{Change in the Monetary Base} = \text{Change in Reserves} + \text{Change in Currency Holdings} = \Delta$$

$$\text{Change in M1} = \text{Change in Deposits} + \text{Change in Currency Holdings} = \frac{\Delta}{\rho + \chi - \rho\chi}$$

At the end, we have balance sheets that look like this:

Central Bank		Private Bank		Household	
Assets	Liabilities	Assets	Liabilities	Assets	Liabilities
Bonds: $b + \Delta$	Reserves: $\rho d + \frac{\rho(1-\chi)}{\rho+\chi-\rho\chi} \Delta$	Reserves: $\rho d + \frac{\rho(1-\chi)}{\rho+\chi-\rho\chi} \Delta$	Deposits: $d + \frac{1-\chi}{\rho+\chi-\chi\rho} \Delta$	Currency: $c + \frac{\chi}{\rho+\chi-\chi\rho} \Delta$	Loans: $\frac{1}{\rho+\chi-\chi\rho} \Delta$
	Currency: $c + \frac{\chi}{\rho+\chi-\chi\rho} \Delta$	Bonds: $B - \Delta$		Deposits: $d + \frac{1-\chi}{\rho+\chi-\chi\rho} \Delta$	
	Net Worth $b - \rho d - c$	Loans: $L + \frac{1}{\rho+\chi-\chi\rho} \Delta$	Net Worth $\rho d + B + L - d$		Net Worth $c + d$

so now the private banks are meeting the reserve requirement exactly:

$$\frac{\text{Reserves}}{\text{Deposits}} = \frac{\rho d + \frac{\rho(1-\chi)}{\rho+\chi-\rho\chi} \Delta}{d + \frac{1-\chi}{\rho+\chi-\chi\rho} \Delta} = \rho$$

The quantity:

$$\omega \equiv \frac{\text{Change in M1}}{\text{Change in the Monetary Base}} = \frac{1}{\rho + \chi - \chi\rho} \quad (10.3.1)$$

is known as the M1 money multiplier. It's called a multiplier because whenever the Central Bank changes the monetary base (which it controls directly), the magnitude of the change in M1 is the change in the monetary base times the multiplier.

The money multiplier depends on two things:

- ρ : the ratio of reserves to deposits ρ . For the most part, the Central Bank can change this number by changing the legal reserve requirement.⁷
- χ : the fraction of their M1 money that people want to hold in physical cash. This is not under the control of the Central Bank: it can vary over time depending of the evolution of payment systems or the public's confidence in the banking system.

In most of what we'll do later, we'll just say that the Central Bank controls the money supply. What we mean by this is that the Central Bank directly controls the monetary base. By understanding how the money multiplier works, it can control M1 fairly accurately, albeit indirectly.

What if the Interest Rate is Zero?

The analysis above was built on the assumption that banks try to maintain reserves as low as possible. This makes sense as long as reserves pay zero interest and other assets pay positive interest. However, if the interest rate fell all the way down to zero (or if the Central Bank started paying market interest rates on reserves), this logic would break down. Banks would be perfectly willing to hold reserves above the legal requirement since the alternatives are not better. This would mean that changes in the monetary base need not lead to changes

⁷As mentioned before, this assumes that banks are trying to keep the minimum possible level of reserves so that the legal requirement is binding.

in the M1 money supply, since all the new reserves would just sit in bank balance sheets without triggering an expansion in loans and deposits.

This scenario has been realized in recent years, as shown in Figure 10.3.1. Nominal interest rates fell to almost zero in late 2008. At around the same time, the Federal Reserve decided to start paying interest on excess reserves (i.e. reserves above the legal reserve requirement). Reserves were suddenly a more attractive asset for banks to hold, and banks started holding large amounts excess reserves. As a result, the M1 money multiplier fell from about 2 to less than 1. The monetary base was increased almost fivefold but M1 increased much less. We'll think more about what happens when the interest rate is near zero in Chapter 15.

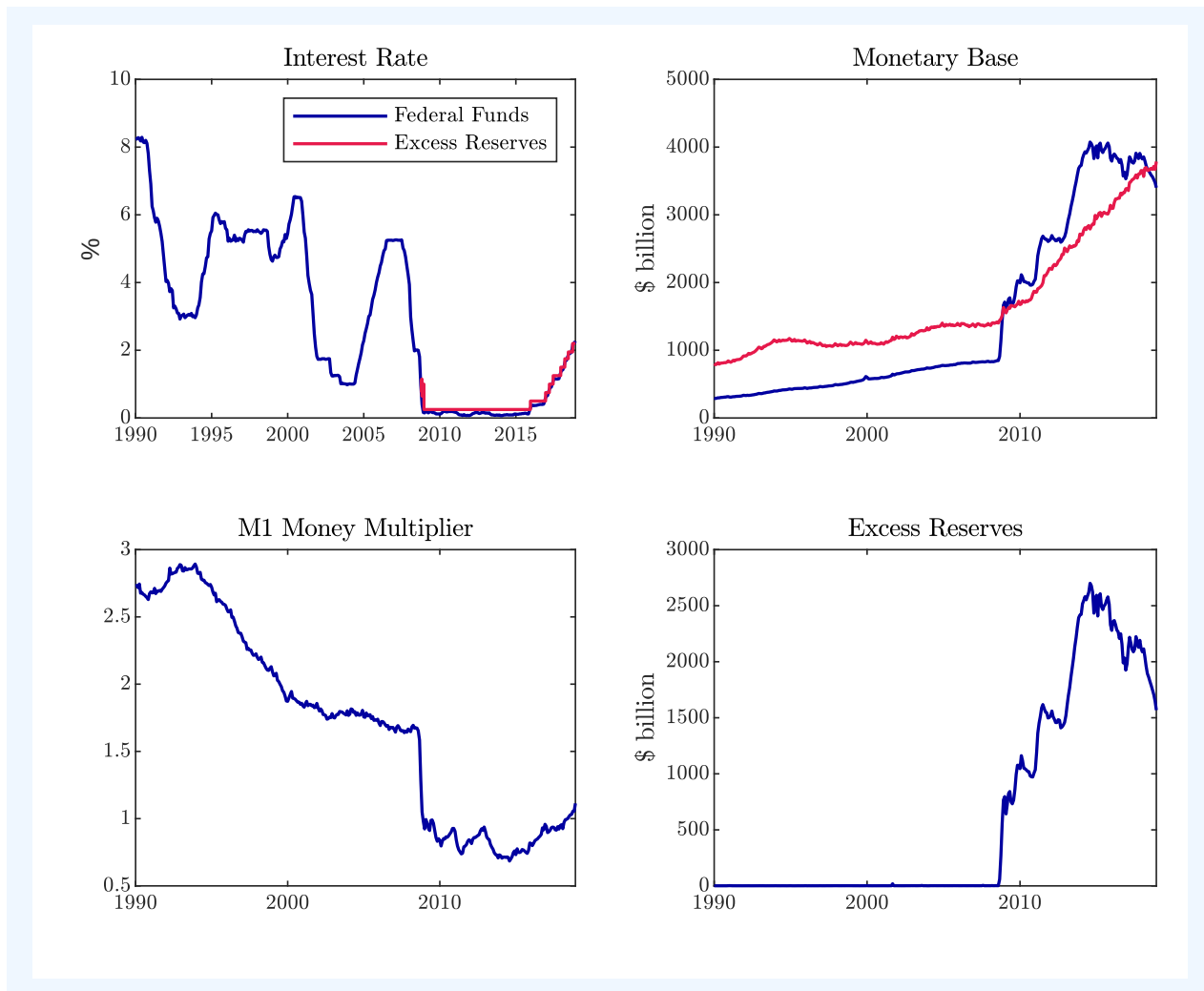


Fig. 10.3.1: Monetary Aggregates in the US when nominal interest rates reached zero. Source: Board of Governors of the Federal Reserve System.

10.4 The Demand for Money

Let's focus on M1 as our definition of money. Why do people hold money? Why is it that instead of only holding assets that earn interest (like physical capital, government bonds, etc.) people choose to hold physical currency and checking deposits, which earn no interest?⁸ We'll propose a simple model based on the idea that money is necessary to carry out transactions. This model is known as the Baumol-Tobin model since it was first analyzed by Baumol (1952) and Tobin (1956).

Let's imagine that there are two types of assets:

- M1 money, which does not pay any interest,
- assets that do pay interest, all of which pay the same nominal interest rate i .

In the course of a period (for instance, the period can be a year), a household will spend Y in real terms (Y stands for real GDP). The price of a good is p , so in nominal terms the household will spend pY . This spending is not all at once: it's spread evenly over the period. For instance, if the period is a year, the household spends $\frac{pY}{365}$ each day.

Whenever the household wants to pay for something, it must use money. However, this does not mean that the household needs to have pY dollars all at once. Whenever it wants, the household can "go to the bank" and swap some of its interest-bearing assets for money. One way of "going to the bank" is to go to the ATM and get physical currency (which is money) from one's savings account (which is not money under the M1 definition). But "going to the bank" need not mean literally going to a physical bank branch. Another way of "going to the bank" is to go to their online brokerage account and sell some bonds (which are not money), depositing the proceeds in a checking account (which is money under the M1 definition). We are going to assume that there is a fixed cost F (in real terms) of going to the bank. F can literally represent ATM fees but also the time and mental cost of dealing with the issue.

The problem of the household is to decide how many times per period it goes to the bank. The advantage of going to the bank many times is that it allows the household to have very low levels of money, so that most of the household's wealth is earning interest most of the time. The disadvantage is that it requires paying the fixed cost F many times. Let N denote the number of times per period that the household goes to the bank. Figure 10.4.1 shows how the amount of money held by the household evolves over time for two values of N .

Each time the household goes to the bank, it brings up its money balance to $\frac{pY}{N}$. Then the balance starts to decrease gradually as the household spends the money. Eventually, when the balance reaches zero, the household goes to the bank again to get more money. It's clear from the picture that the household will, on average, hold less money the more often it goes to the bank. Indeed, the average money balance is simply:

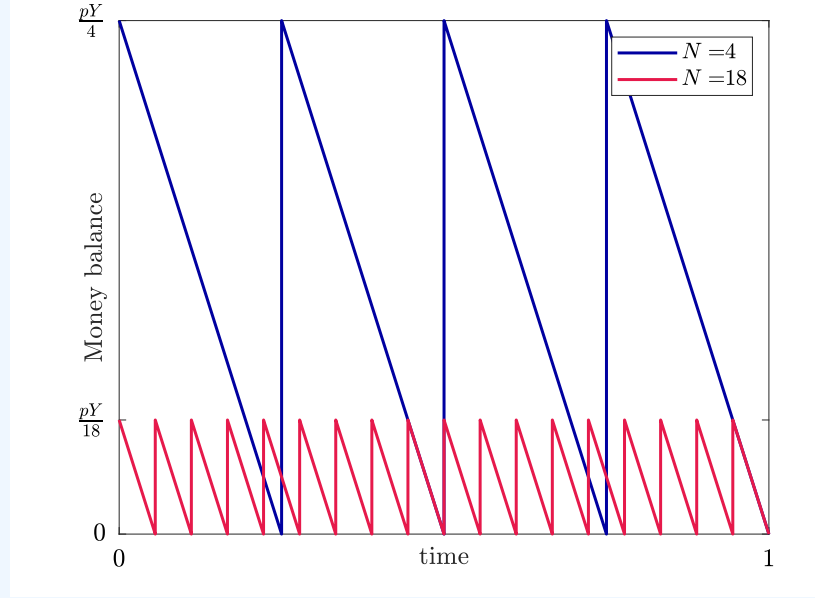
$$M = \frac{pY}{2N} \quad (10.4.1)$$

How does the household choose N ? Mathematically, it solves the following problem:

$$\min_N pFN + i\frac{pY}{2N} \quad (10.4.2)$$

⁸Some checking deposits do earn interest, but it's typically much lower than what one could earn by holding some other asset.

Fig. 10.4.1: Money balances over time in the Baumol-Tobin model.



What does this mean? The household is trying to minimize the overall cost of having money for transactions. This cost has two parts. First, if it goes to the bank N times, it pays the cost F each time. Expressed in nominal terms, this gives us pFN . Second, if it goes to the bank N times it will on average hold $\frac{pY}{2N}$ dollars in money. Since this money does not earn interest, there is an opportunity cost of holding it: the foregone interest that the household could have earned if it had held less money. If the interest rate is i , then $i\frac{pY}{2N}$ is the foregone-interest cost of the household's money holdings.

The first-order condition for problem (10.4.2) is:

$$pF - \frac{ipY}{2} N^{-2} = 0$$

so we can solve for N to get:

$$N = \sqrt{\frac{iY}{2F}} \quad (10.4.3)$$

Equation (10.4.3) tells us that the household will go more times to the bank if i is high and if F is low. What's the economic logic of this? If i is high, then the opportunity cost of holding money is high and the household will be willing to go to the bank many times in an effort to hold low amounts of money. On the other hand, if F is low, going to the bank is cheap and the household will, other things being equal, be willing to go to the bank more times.

Replacing (10.4.3) into (10.4.1) and rearranging, we get an expression for the average money balances:

$$M = p\sqrt{\frac{YF}{2i}}$$

or, dividing by the price level, for average “real” money balances:

$$\frac{M}{p} = \sqrt{\frac{YF}{2i}} \quad (10.4.4)$$

“Real money balances” are the answer to the question: “how many goods would the household be able to buy with the amount of money it holds?” Equation (10.4.4) is telling us that real money balances will be higher when:

- Y is high. If the household wants to spend more, this will involve more payments and therefore the household will choose to carry higher real money balances.
- i is low. i is the opportunity cost of holding money. If this is low, the household will choose to hold higher money balances to save on trips to the bank.
- F is high. If going to the bank is costly, the household will choose to hold higher money balances to save on trips to the bank.

Figure 10.4.2 shows the shape of the money-demand function that results from the Baumol-Tobin model. The quantity of money that people want to hold decreases with the interest rate. A higher cost of going to the bank shifts the entire demand schedule to the right.

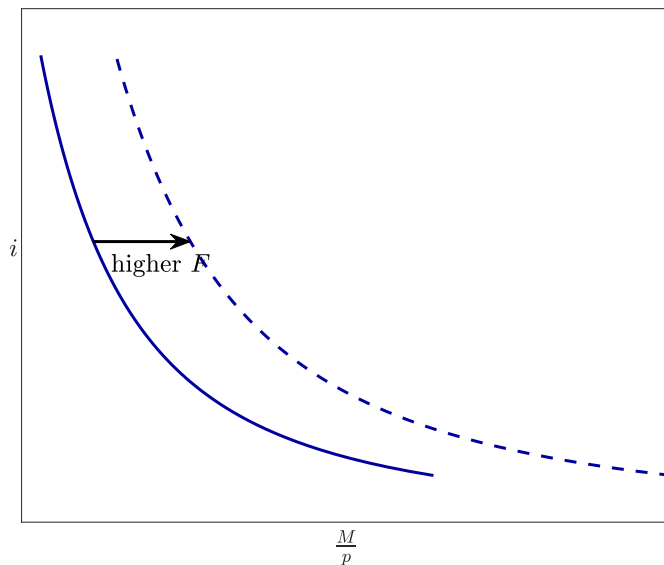


Fig. 10.4.2: The money demand function in the Baumol-Tobin model.

The Baumol-Tobin model makes very specific assumptions about how exactly households manage their money: every trip to the bank costs the same, spending is spread out exactly over time and perfectly predictable, etc. We will sometimes want to think about the basic economic forces that the Baumol-Tobin model captures while not expecting the exact formula (10.4.4) to hold. For this purpose, we will sometimes want to

think of a generalized money-demand function $m^D(Y, i)$, increasing in Y and decreasing in i . $m^D(Y, i) = \sqrt{\frac{YF}{2i}}$ is just a special case of this more general formula.

There is some debate as to whether the money demand function is sufficiently stable over time to be a useful thing to look at. At times, the quantities of M0, M1 and M2 that people held have moved around quite a bit without changes in interest rates or GDP that would account for them. However, Lucas and Nicolini (2015) and Kurlat (2019) argue that if one constructs an appropriately-weighted composite measure that takes into account how people substitute between different subcomponents of money like physical currency, checking accounts, savings accounts, etc., the resulting money demand has a relatively stable relationship with the interest rate, as predicted by the theory. This is illustrated in Figure 10.4.3.

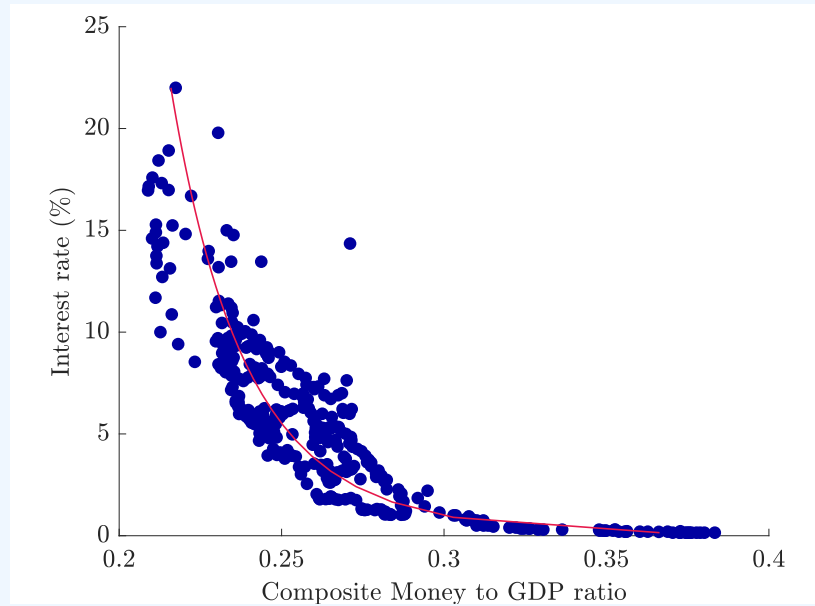


Fig. 10.4.3: Money demand in the United States. Each dot represents one month between 1980 and 2013. Source: Kurlat (2019)

Exercises

10.1 Central Bank Instruments

Suppose the Central Bank wants to reduce the M1 money supply but does not want to change the monetary base. In what direction should it change reserve requirements?

10.2 Pickpockets

Suppose there is an increase in the number of pickpockets. How would that change the fraction of their money that people want to have in cash as opposed to checking deposits? If the Central Bank keeps the monetary base constant, what will happen to the M1 money supply?

10.3 Interest on Reserves

Consider an economy where households almost use no cash (so $\chi \rightarrow 0$). Furthermore, assume that the Central Bank makes lowers the reserve requirement to almost zero (so $\rho \rightarrow 0$).

- (a) What is the value of the money multiplier as χ and ρ become small?
- (b) Suppose that the Central Bank decides to pay interest on reserves at the market rate i . Does equation (10.3.1) still hold? Why or why not?
- (c) Under this policy regime, what determines the quantity of money?

10.4 ATMs

Suppose one day, suddenly and unexpectedly, ATMs are invented, which make getting cash more convenient than before. What would be the effect of this invention on the demand for money?

10.5 Going to the Bank

The average person in the US has about \$10,000 in M1 money and spends around \$60,000 per year. Suppose this person uses its checking account to pay for all its purchases, behaves according to the Baumol-Tobin model and only has two investment options: its checking account (which pays zero interest) and an investment account (which pays interest). Define “going to the bank” as transferring a balance from the investment account to the checking account.

- (a) How often does the average person go to the bank?
- (b) Suppose the interest rate is 2%. What must be the perceived cost of each visit to the bank for this type of behavior to be optimal?
- (c) How often would the average person go to the bank if the interest rate rose to 3%?
- (d) Now suppose the checking account does pay interest, but it only pays 0.5% instead of the full 2% one can get from the investment account. What is the opportunity cost of holding balances in the checking account? What do we infer now about the perceived cost of going to the bank?

CHAPTER 11

The Price Level and Inflation

11.1 Measurement

Inflation is defined as a generalized increase in the level of prices. If the prices of all goods increased by the same percentage, then measuring inflation would be straightforward. It becomes harder when different prices are changing at different rates, or even going in different directions. How do we define the “overall” level of change?

We already encountered this issue when we discussed real and nominal GDP in Chapter 1. There the question was how to measure the “overall” change in output when prices of different goods were changing by different percentages. Here we are interested in prices for their own sake.

The basic idea is going to be to define what is known as a “basket” of goods (i.e., a list of specific quantities of various goods) and measure how the total price of the basket changes. We call this total price a price index. The different methods of measuring the total change in prices have to do with different ways of choosing and updating the basket of goods.

The GDP Deflator

The GDP deflator is a price index that is a side product of the calculation of real GDP. It is defined as

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100.$$

Look back at Example 1.13 from Chapter 1. Suppose that Expandia computes real GDP at 2017 prices, so that real GDP in 2018 is \$2,550, while nominal GDP is \$1,860. Then the GDP deflator would be

$$\text{GDP deflator}_{2018} = \frac{\$1,860}{\$2,550} \times 100 \approx 73.$$

By definition, the GDP deflator is 100 in the base year, so in this case we would say that overall prices went down. Notice that not all prices went down: some went up and some went down. By using the GDP deflator as a price index we are implicitly choosing to weigh each good in proportion to its share of GDP.

The Consumer Price Index

The most commonly used price index weighs the prices of different goods by how much they are consumed rather than how much they are produced. The basket for the CPI is constructed by conducting a survey asking households how much they consume of each good. The CPI is then defined by:

$$CPI = \frac{\sum_j q_j p_{jt}}{\sum_j q_j p_{j0}} \times 100$$

where:

- q_j is the quantity of good j in the basket,
- p_{jt} is the price of good j in period t ,
- p_{j0} is the price that good j used to have in the base year.

Example 11.1.

The residents of Luxuria consume only three goods: Ferraris, caviar, and champagne.

Good (j)	Quantity (q_j)	Price in 2017 (p_{j0})	Price in 2018 (p_{jt})
Ferrari (units)	2	100	115
Caviar (kg)	20	4	3
Champagne (liters)	10	2	4
Total basket	1	300	330
CPI		100	110

Inflation

Having measured a price index, we calculate inflation (denoted by the letter π) by applying the following formula:

$$\pi_t = \frac{P_t}{P_{t-1}} - 1$$

where P_t is a price index in period t . In Example 11.13, inflation (in terms of the GDP deflator) was -27% . When inflation is negative we call it deflation: a general fall in prices. In Example 11.1, inflation (in terms of the CPI) was 10% .

Notice that for a given country there will coexist several measures of inflation, each derived from a different price index. Usually it doesn't make much difference which price index one looks at because the production-based basket that is used in constructing the GDP deflator and the consumption-based basket that is used in constructing the CPI are not that different, at least in the US. It could make a bigger difference in countries that produce and consume very different goods. For instance, in a country that produces oil and exports most of it, a rise in the price of oil would result in a big rise in the GDP deflator but not as much in the CPI.

Figure 11.1.1 shows the evolution of CPI inflation in the US. Inflation was very variable until the 1950s, with times of over 20% inflation and over 10% deflation. Between the 1960s and the early 1980s inflation tended to increase. Since the mid-1980s inflation has been quite low and stable.

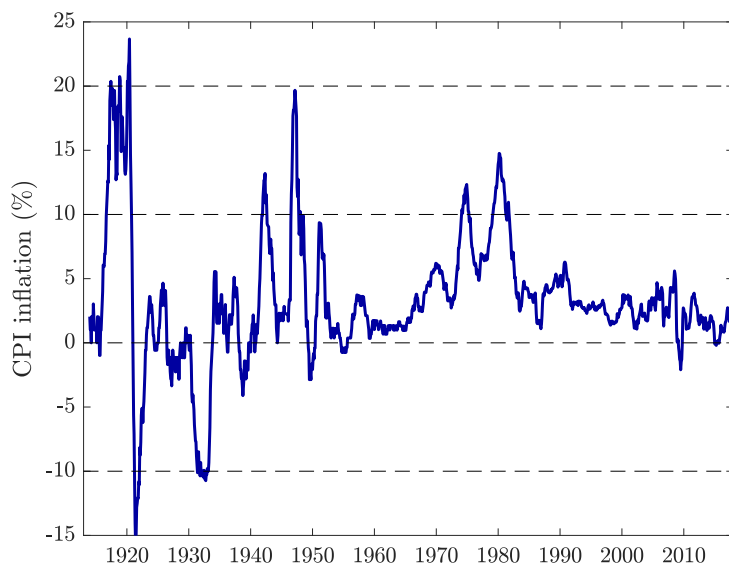


Fig. 11.1.1: CPI inflation in the US. Source: BLS.

Nominal and Real Interest Rates

A basic lending transaction works as follows:

- a lender gives a borrower one dollar in period t ,
- the borrower pays back $1 + i_{t+1}$ dollars in period $t + 1$.

i_{t+1} is the interest rate between periods t and $t + 1$.¹

We will often talk about “the” interest rate, although in reality there is no single interest rate for all loans. Typically, the interest rate on government debt is the lowest rate in the country (at least in the US, where the government is perceived as reliable) and rates paid by private borrowers are higher, which compensates for administrative costs, the probability of default, etc.

We are often interested in expressing interest rates in terms of goods rather than in terms of dollars.

Example 11.2.

The interest rate in Usuria on a one year loan that is issued in January 2018 and will be paid back in

¹An interest rate always involves more than one period: when the loan starts and when it ends. We will adopt the convention to label interest rates according to the period when the loan has to be paid back. Hence i_{t+1} refers to the interest rate on loans that are issued in period t and are due in period $t + 1$.

January 2019 is 11%. Everyone expects that inflation between those dates will be 2%. Suppose someone lends 100 dollars in January 2018. What are they giving up? What do they get in return?

	January 2018	January 2019
Price index	100	102
Loan issued / repayment received (in dollars)	100	111
Loan issued / repayment received (in goods)	$\frac{100}{100} = 1$	$\frac{111}{102} \approx 1.088$

In the example, the 100 dollars of the original loan would be enough to buy exactly 1 consumption basket at the time the loan is granted. By the time the loan is repaid, the 111 dollars that are paid back are not enough to buy 1.11 consumption baskets because prices have risen in the meantime: it is only enough to buy 1.088 consumption baskets. In other words, for each good that the lender gave up at the beginning, he is getting back 1.088 goods one year later. The 0.088 extra goods that the lender obtains are what we call a real interest rate. We call it “real” because it is expressed in terms of goods as opposed to a “nominal” rate that is expressed in dollars. Whenever we have referred to interest rates in Chapters 4-9 we have meant real interest rates because we were thinking about exchanges of real goods over time. Instead, when we studied money demand in Chapter 10 it was the nominal interest rate that mattered because that’s what determines the opportunity cost of holding money.

In general, if i_{t+1} is the nominal interest rate, the real interest rate is defined by the following expression:

$$\begin{aligned}
 1 + r_{t+1} &= \frac{\text{Goods you can afford with loan repayment}}{\text{Goods you could afford with the loan when issued}} \\
 &= \frac{\left(\frac{1+i_{t+1}}{P_{t+1}}\right)}{\frac{1}{P_t}} = \frac{1+i_{t+1}}{\frac{P_{t+1}}{P_t}} \\
 &= \frac{1+i_{t+1}}{1+\pi_{t+1}} \\
 \Rightarrow r_{t+1} &\approx i_{t+1} - \pi_{t+1}
 \end{aligned} \tag{11.1.1}$$

(The last approximation is accurate when π_{t+1} is small.) Equation (11.1.1) is known as the Fisher equation, named after Irving Fisher.

It is not always easy to know what the real interest rate is. There is always some uncertainty as to what inflation is going to be. If one lends or borrows in dollars, as is usual, then until the end of the loan one is not certain how many goods the future dollars are going to be worth. Sometimes we make the distinction between ex-ante real interest rates (meaning the real rate that was expected at the beginning, based on expected inflation) and ex-post real interest rates (based on what inflation turned out to be).

11.2 Equilibrium in the Money Market

In Chapter 10 we looked at the money supply and the money demand separately. An equilibrium in the money market requires that supply equals demand: all the money that is created jointly by the Central Bank and the private banks must be held by someone, voluntarily. We can write the money-market equilibrium condition

as:

$$M^S = m^D(Y, i) \cdot p \quad (11.2.1)$$

The left hand side of (11.2.1) is the money supply. We are going to imagine that the Central Bank simply chooses the money supply, by choosing the monetary base and understanding the money multiplier. The right hand side of (11.2.1) is the money demand. This is the result of households' decisions of how much money to hold.

How does a money market equilibrium come about? Suppose that the central bank increases M^S , what changes to induce households to increase their money holdings? The right hand side of (11.2.1) gives us a list of the things that could possibly change to restore equilibrium:

- p . The price level could rise. If the price level is higher, then the same amount of real transactions requires more money, so households would want to hold extra money.
- i . Nominal interest rates could fall. If interest rates are lower, the opportunity cost of money is lower and households would be willing to hold more of it.
- Y . GDP could go up. If GDP is higher, there are more real transactions to carry out, which requires more money.

There are different views on which of these three variables tends to respond and why. This turns out to be an extremely important issue. In this chapter we'll look at the so-called "classical" view, which postulates that the real side of the economy is separate from anything having to do with money. Real variables like real GDP and real interest rates are determined purely by real factors (technology, preferences, etc.) that do not change when the money supply changes. One way of stating this view is to say that money is *neutral*. In everything we have done so far we have implicitly adopted this classical view: we studied the forces that determine real variables without any reference to the money supply. In Chapter 14 we'll think about reasons why money might *not* be neutral. For now, let's see how prices and inflation behave if the classical view is correct.

An Economy in Steady State with a Constant Money Supply

Imagine first that the economy is in a steady state where Y and r are constant and the Central Bank holds the money supply M^S constant as well. We'll conjecture that in this economy the price level will be constant as well, and then verify that this is consistent with an equilibrium in the money market.² If indeed the price level is constant, then the nominal interest is equal to the real interest rate. Therefore, solving for p in (11.2.1), we get:

$$p = \frac{M^S}{m^D(Y, r)} \quad (11.2.2)$$

which indeed is constant, because we have assumed that M^S , Y and r are constant. Equation (11.2.2) tells us that an economy where the money supply is higher will, other things being equal, have higher prices.

²This method of figuring out the equilibrium of a model is sometimes called "guess and verify". Technically, we will show that there is an equilibrium where prices are constant but not that it's the only equilibrium.

People want a certain level of real money balances given by $m^D(Y, r)$, so the price level will be such that $\frac{M^S}{p}$ corresponds to these desired real money balances.

An Economy in Steady State with a Growing Money Supply

Maintain the assumption that Y and r are constant but now assume that the money supply grows at a constant rate μ , i.e. $M_{t+1}^S = (1 + \mu) M_t^S$. We'll conjecture that in this economy the price level will also grow at rate μ , and then check that this is consistent with equilibrium in the money market. If:

$$p_{t+1} = (1 + \mu) p_t$$

then inflation π_{t+1} is:

$$\pi_{t+1} \equiv \frac{p_{t+1}}{p_t} - 1 = \mu$$

and therefore the nominal interest rate is:

$$i_{t+1} = r + \pi_{t+1} = r + \mu$$

If the money market is in equilibrium in period t , then:

$$\begin{aligned} M_t^S &= m^D(Y, r + \mu) p_t \\ \Rightarrow M_t^S (1 + \mu) &= m^D(Y, r + \mu) p_t (1 + \mu) \\ \Rightarrow M_{t+1}^S &= m^D(Y, r + \mu) p_{t+1} \end{aligned}$$

which implies that the money market is also in equilibrium in period $t + 1$. This confirms our conjecture.

Economically, what's going on is the following. Since GDP and nominal interest rates are constant, people want to hold constant real money balances. Since the money supply is growing, prices must be growing too in order to keep money balances constant.

Figure 11.2.1 looks at data on inflation and the growth rate of the money supply over a long period in many countries. Comparing across countries, the data shows that inflation is almost exactly proportional to the growth rate of the money supply.

A Growing Economy

Now suppose that the economy is in a steady-state-with-growth, with Y growing at a constant rate g and a constant real interest rate r . The money supply grows at a constant rate μ . Let's try to find the inflation rate in this economy. Start from (11.2.1) and take the derivative with respect to time:

$$\frac{dM^S}{dt} = \left[\frac{\partial m^D(Y, i)}{\partial Y} \frac{dY}{dt} + \frac{\partial m^D(Y, i)}{\partial i} \frac{di}{dt} \right] \cdot p + m^D(Y, i) \frac{dp}{dt}$$

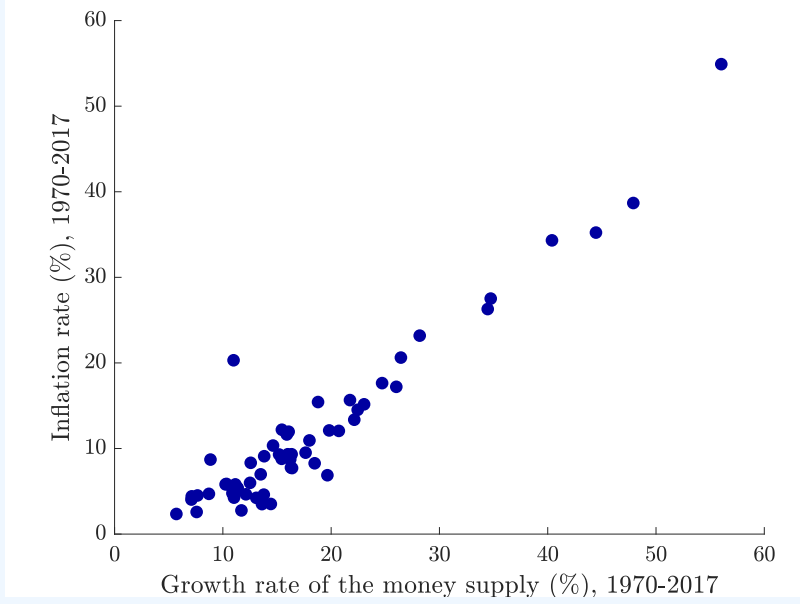


Fig. 11.2.1: Inflation and the growth rate of the money supply in the long run. Each dot represents one country. Source: World Bank.

Now divide by (11.2.1) on each side:

$$\frac{\frac{dM^S}{dt}}{M^S} = \left[\left(\frac{\partial m^D(Y, i)}{\partial Y} \frac{Y}{m^D(Y, i)} \right) \frac{\frac{dY}{dt}}{Y} + \frac{\frac{\partial m^D(Y, i)}{\partial i} \frac{di}{dt}}{m^D(Y, i)} \right] + \frac{\frac{dp}{dt}}{p}$$

$$\mu = \left[\left(\frac{\partial m^D(Y, i)}{\partial Y} \frac{Y}{m^D(Y, i)} \right) g + \frac{\frac{\partial m^D(Y, i)}{\partial i} \frac{di}{dt}}{m^D(Y, i)} \right] + \pi$$

Let $\eta \equiv \frac{\partial m^D(Y, i)}{\partial Y} \frac{Y}{m^D(Y, i)}$. η represents the elasticity of money demand with respect to GDP. It is the answer to the question: if GDP rises $x\%$, by what percent does the demand for real money balances increase? Assume the function m^D is such that this elasticity is constant, so:

$$\mu = \left[\eta g + \frac{\frac{\partial m^D(Y, i)}{\partial i} \frac{di}{dt}}{m^D(Y, i)} \right] + \pi$$

If inflation is constant, then $i = r + \pi$ will be constant so $\frac{di}{dt} = 0$. Then the equation reduces to:

$$\mu = \eta g + \pi$$

and therefore:

$$\pi = \mu - \eta g \quad (11.2.3)$$

so, indeed, inflation is constant. Equation (11.2.3) tells us that, other things being equal, a growing economy will have lower inflation. Why is this? A growing economy means a growing number of transactions and

therefore a growing demand for real money balances. This means that the economy can absorb growing quantities of money without resulting in inflation. Why does η show up in the formula? η measures *how much* the demand for money increases when the economy grows. The higher this number, the faster the money supply can grow without leading to inflation. Note that formula (11.2.3) encompasses the steady-state-without-growth examples as special cases.

A One-Time Increase in the Money Supply

Suppose that, starting from a steady state with a constant money supply, at time t there is a sudden, unexpected increase in the money supply, from M^S to $M^{S'}$. After this, the money supply is expected to remain constant at $M^{S'}$ forever. What's going to happen to the price level?

Before time t we had that $p = \frac{M^S}{m^D(Y,r)}$. After that, the money supply will again be constant, except that the level will be higher. Therefore we are going to be back in a constant-money-supply steady state, where $p' = \frac{M^{S'}}{m^D(Y,r)}$. The effect on prices is therefore:

$$\frac{p'}{p} = \frac{M^{S'}}{M^S}$$

In other words, prices jump immediately to their new level, and the size of the jump is proportional to the size of the increase in M^S .

The exact causal chain that leads from an increase in the money supply to a rise in the price level is a matter of some debate. Why does everyone immediately raise their prices? In terms of the logic of the model, what happens is that when the Central Bank increases the money supply, everyone suddenly has more money than they would like, so they try to reduce their money balances. But it is impossible for everyone to do this at the same time because someone has to hold the money. Everyone immediately realizes what's going on so money immediately loses value, which is exactly what an increase in the price level means. Note that one condition for this reasoning to be correct is that prices must be flexible, reacting immediately to changes in the supply of money. Starting in Chapter 14, we'll think about the possibility that prices might be "sticky" and react slowly to changes in the money supply. This will be a source of monetary nonneutrality, i.e. of interaction between money and the real economy.

A Change in the Rate of Growth of the Money Supply.

Now let's do a slightly more subtle exercise. Suppose we start at a steady state with the money supply growing at rate μ and, therefore, an inflation rate of μ . At time t there is a sudden, unexpected increase in the *rate of growth* of the money supply, from μ to μ' . After this, the rate of growth of the money supply is expected to remain at μ' forever. What's going to happen?

A naive guess would be to say that inflation will simply increase from μ to μ' . This guess is not wrong, but it's incomplete. If the inflation rate changes from μ to μ' , then the nominal interest rate rises from $i = r + \mu$ to $i = r + \mu'$. Using (11.2.1), this implies that real money balances must fall. Higher nominal interest rates increase the opportunity cost of holding money, so people want to hold less of it. But the *level* of M^S does

not change at time t : it simply starts growing at a different rate. What makes the money market clear? The price level must rise!

Economically, this is what's going on. People are all simultaneously trying to reduce their money balances because the opportunity cost of holding them has gone up. Since the total (nominal) supply of money has not changed, money loses value, which is the same thing as saying the prices rise.

Figure 11.2.2 shows how the price level evolves over time in the different examples above.

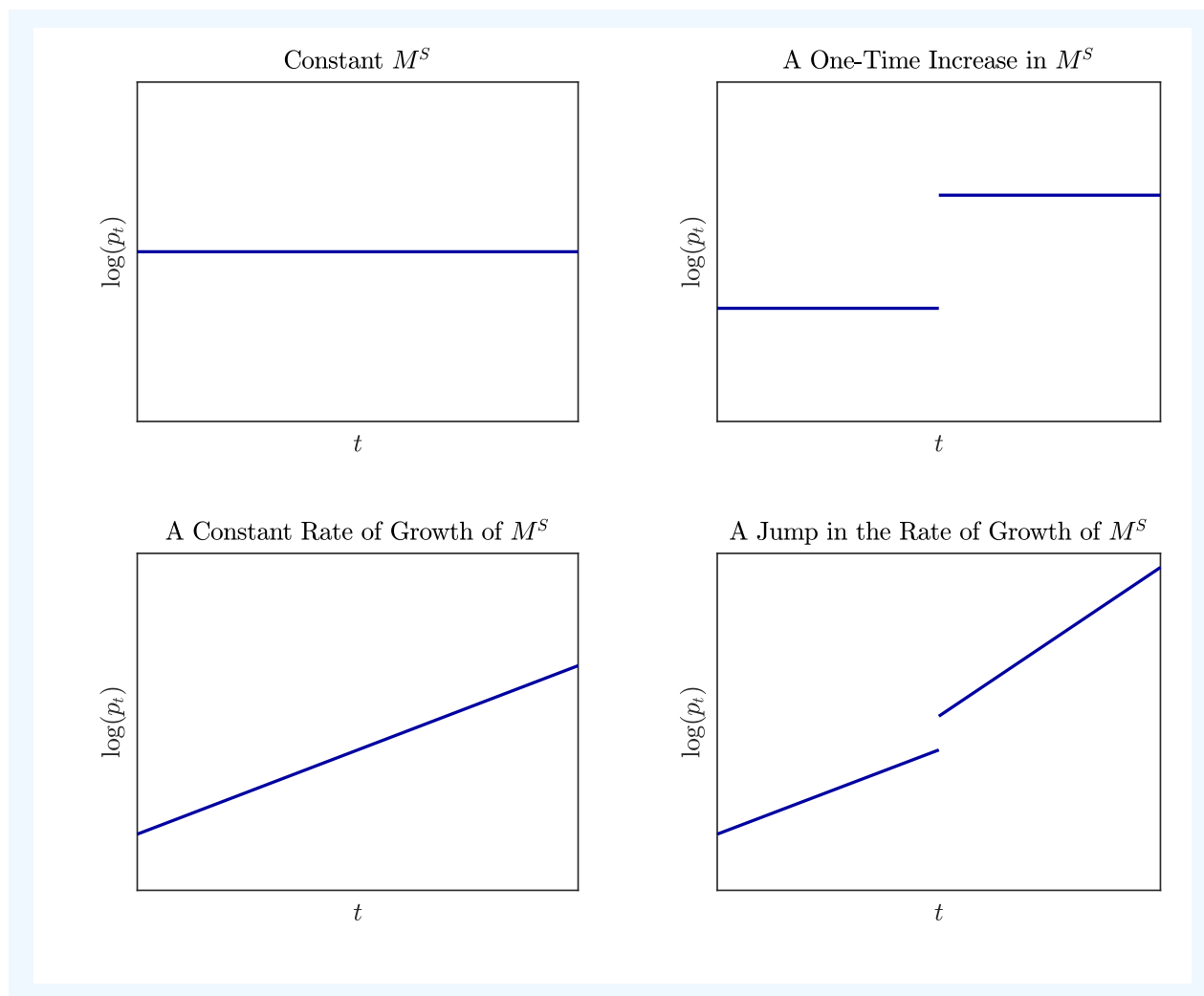


Fig. 11.2.2: The evolution of prices in several examples.

The Velocity of Money

The “velocity” of money refers to the number of times a unit of money is used per period. Let's see an example.

Example 11.3.

The money supply is \$2. At the beginning of the period, Ann and Bob each hold one dollar. Over the course of the year, the following things happen:

- Ann produces an apple and sells it to Bob for \$1. Bob produces a banana and sells it to Ann for \$1
- Ann produces asparagus and sells it to Bob for \$1. Bob produces a blueberry and sells it to Ann for \$1
- Ann produces an apricot and sells it to Bob for \$1. Bob produces a blackberry and sells it to Ann for \$1

In the example, nominal GDP is \$6, the money supply is \$2 and each dollar changes hands 3 times, so the velocity of money is 3. In general, we have that:

$$\underbrace{M}_{\text{Money Supply}} \underbrace{V}_{\text{Velocity}} \equiv \underbrace{\underbrace{p}_{\text{Price Level}} \underbrace{Y}_{\text{Real GDP}}}_{\text{Nominal GDP}} \quad (11.2.4)$$

Equation (11.2.4), sometimes known as the quantity equation, is a *definition*. It's true because this is the way we define the velocity of money.

How does equation (11.2.4) relate to the money-market equilibrium condition (11.2.1)? We can use (11.2.1) to replace $\frac{M}{p}$ in (11.2.4) and rearrange to obtain:

$$V = \frac{Y}{m^D(Y, i)} \quad (11.2.5)$$

Equation (11.2.5) says that any theory of money demand, summarized by a function $m^D(Y, i)$, is also a theory of velocity. Once we have a $m^D(Y, i)$ function, we can simply plug it into (11.2.5) to obtain velocity as a function of Y and i .

Our theory of money demand implies that velocity is an increasing function of the nominal interest rate. We can see this in equation (11.2.5) by noting that m^D is decreasing in i , which implies V is increasing in i . Economically, what this is saying is that if interest rates are higher, people will hold less money, so in order to carry out the same amount of real transactions, each dollar will have to change hands more times.

Figure 11.2.3 shows how the velocity of money has evolved over time. Notice that the velocity of M1 is higher than the velocity of M2. Recall from the definitions of M1 and M2 that M2 includes more things than M1. Using (11.2.4), this implies that the velocity of M2 must be lower.

One assumption that people sometimes make is that V is constant. Figure 11.2.3 shows that this is not completely justified, since velocity has moved around quite a bit over time. Furthermore, a standard model of the money demand says that velocity should not be expected to remain constant: when interest rates rise, money demand falls and therefore velocity rises.

Nevertheless, sometimes it is useful to assume that velocity can be held constant in an “other things being

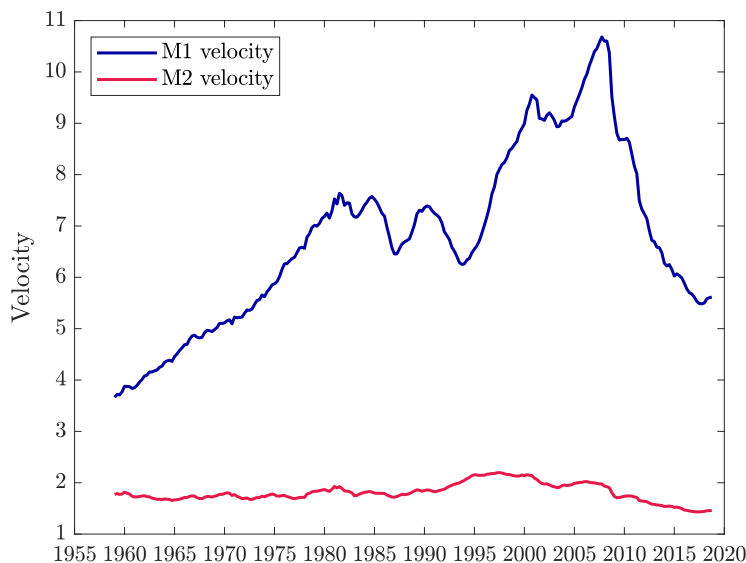


Fig. 11.2.3: The velocity of money in the US, using M1 and M2 as definitions of money. Source: Federal Reserve Bank of St. Louis.

equal” sense when one considers some other change in the economic environment. If one assumes that V is constant, then equation (11.2.4) changes from being a *definition* to being a *theory*. In fact, it is sometimes known as the “quantity theory”, because it says that the price level will be exactly proportional to the quantity of money.

The evidence from Figure 11.2.1 is sometimes interpreted as supportive of the quantity theory, since the quantity theory implies that there should be an exact linear relationship between changes in the money supply and changes in prices. Notice that the relationship between money growth and inflation becomes much closer for countries with high inflation. Even if velocity is not exactly constant, compared to the scale of changes in the money supply in those countries it doesn’t move that much, so the quantity theory is not such a bad approximation.

11.3 Seignorage

Nowadays, most countries tend to keep inflation quite low, though usually not at zero. We’ll look at some of the arguments in favor of positive inflation later on. Historically, one of the reasons why inflation has sometimes been high is that governments used inflation to obtain seignorage.

The term seignorage, which derives from the French word for lord (“seigneur”), originally referred to the profit made in the production of coins. Back when coins were usually made of precious metals, the value of a minted coin was typically above the value of the metal used to produce it. Why? Because minted coins were better money than raw metal since they were standardized to be useful in transactions. The profit earned by the mint by turning metal into money was known as seignorage. Nowadays the term is used more broadly to refer to the resources obtained thanks to the ability to create money.

When we looked at the process of money creation in Chapter 10, we didn't pay too much attention to the Central Bank's balance sheet, but if you go back to it, you'll notice that, in the process of increasing the monetary base by Δ , the Central Bank increased both its assets and its liabilities by Δ . This doesn't seem like a big deal, but there is one important difference. The assets that the Central Bank obtains (government bonds) earn interest while the liabilities that it issues (currency and reserves) typically do not. Standard accounting still treats them as liabilities but in a certain sense they are not. Furthermore, the Central Bank is just a branch of the government.³ Therefore increasing the monetary base is a way for the government to get a loan that it will never have to repay and doesn't pay any interest. Indeed, one way to write down the government's budget constraint is:

$$B_{t+1} + p_t \tau_t + [M_{t+1}^B - M_t^B] = p_t G_t + (1 + i_t) B_t \quad (11.3.1)$$

where:

- B_{t+1} is nominal public debt,
- G_t is real government spending,
- τ_t is real government tax revenue,
- p_t is the price level,
- i_t is the nominal interest rate,
- M_t^B is the monetary base.

Let's go through the terms in (11.3.1) to see what it means. The right hand side represents all the payments the government must make, in nominal terms. $p_t G_t$ is how much the government must pay for the current period's spending. $(1 + i_t) B_t$ is how much it must pay on the debts it had at the beginning of the period, including the interest that accrued in the current period. The left hand side represents all the resources the government can use to make its payments. $p_t \tau_t$ is how much it raises in taxes, in nominal terms. $M_{t+1}^B - M_t^B$ is the increase in the monetary base. This is all the payments the government can make just by virtue of having created extra money. B_{t+1} is the amount of payments the government can make by virtue of issuing debt that will have to be repaid in the future.

We can also rearrange (11.3.1) to express it as:

$$M_{t+1}^B + B_{t+1} = p_t [G_t - \tau_t] + (1 + i_t) B_t + M_t^B$$

This formulation makes it easier to see that the monetary base is just like debt (in the sense that it enters the government budget in the same way), except that it doesn't pay interest.

Historically, governments have used expansion of the monetary base as a way to satisfy the government budget in various circumstances. Sometimes it's a result of difficulties in collecting regular taxes, due to tax

³This is literally true in some countries and sort-of-true in others. In the US, the Federal Reserve has a mixed governance structure with some influence from the private sector. However, it rebates its profits to the Treasury, so in that sense it's part of the Federal Government.

evasion or political indecision about what other taxes to use. Sometimes it's a result of a rapid increase in government spending that leaves no time to increase regular taxes, as in wartime. Sometimes it's the result of the inability to borrow, perhaps because lenders don't trust the government to pay back its debts. In other instances it may have been to a misperception that increasing the monetary base is a way for the government to obtain resources without really taking them away from anyone. Often it could be several of these reasons at the same time.

One subtle question is who exactly the government is taxing when it increases the monetary base. It's clear that the government can, at least to some extent, pay for goods and services with monetary expansion. But nobody seems to be paying for this. How does it all add up? The answer is that anyone who holds money is implicitly paying a tax to the government when the monetary base expands. We know that in a money market equilibrium, an expansion of the monetary base leads (through the money multiplier) to an increase in the money supply and then to an increase in the price level. Anyone who holds the monetary base *while it's losing value* is implicitly giving up some of their wealth to the government, just as they would if they were paying a regular tax. That's why seignorage revenue is also sometimes referred to as an inflation tax.⁴

Governments usually limit how much seignorage revenue they try to raise because they want to avoid creating high inflation. But even if they didn't care about inflation there is a limit to how much seignorage revenue they can obtain. To obtain high revenue, they need to expand the monetary base very fast. But fast expansion of the monetary base leads to high inflation, which leads to high nominal interest rates, which means that the real money demand falls. Since implicitly seignorage is a tax on money holdings, this means that the tax base shrinks. Exercise 11.6 asks you to work out the limit on seignorage revenue and compare it to some historical experiences of very high inflation.

11.4 The Cost of Inflation

Inflation is generally seen as undesirable. There are several reasons for this.

One reason can be understood directly from the Baumol-Tobin model of money demand. Other things being equal, more inflation implies higher nominal interest rates, which means that people will "go to the bank" more times to avoid holding high money balances. Each of those trips to the bank has a cost of F .⁵ Using (10.4.3) we can compute the total cost of trips to the bank as:

$$\begin{aligned} \text{Cost} &= NF \\ &= \sqrt{\frac{iYF}{2}} \\ &= \sqrt{\frac{(r + \pi)YF}{2}} \end{aligned} \tag{11.4.1}$$

At the times of high inflation in Argentina in the late 1980s, my dad would literally go to the bank twice a day, once around noon and once after work just to make sure that he had exactly enough money for the day's

⁴Technically, this is not quite right. The government can obtain seignorage revenue with zero inflation if the economy is growing. Exercise 11.3 asks you to work out how much.

⁵Sometimes this is known as "shoe leather cost": people wear out their shoes by walking to the bank all the time.

expenses and no more. That time spent dealing with the problem of how much money to hold has a real opportunity cost.

On the basis of a reasoning like this, Milton Friedman advocated keeping nominal interest rates at or very near zero, a policy known as the “Friedman Rule”. The idea of the Friedman Rule is to eliminate the opportunity cost of holding money. In formula (11.4.1), having $i = 0$ would make the cost equal to zero, because it would mean that you don’t ever need to go to the bank: since it has no opportunity cost, you can just hold all your wealth in money. Notice that in order to have $i = 0$, one would need to have $\pi = -r$. Since the real interest rate is usually positive, this means that implementing the Friedman Rule requires deflation. The Friedman Rule is usually seen as a theoretical extreme, more valuable for the underlying logic than as a concrete policy proposal.

Economists sometimes refer to “menu costs” as part of the cost of inflation. Sometimes there are real resources that need to be dedicated to put in place a change in prices. For instance, restaurants need to print new menus, shops need to print new signs, etc. When inflation is high this needs to be done more often, which is a real cost. Minimizing menu costs would require keeping inflation at zero, rather than running deflation as implied by the Friedman Rule. Even this would not eliminate menu costs: zero inflation means that the price index would stay constant, but the prices of individual goods would still move up and down a lot, and making those changes would incur menu costs. There is some disagreement among economists about whether it’s plausible that menu costs are large.

Another cost of inflation is that it creates uncertainty about relative prices. In order to decide what to buy, people need to know the prices of different goods. But they don’t look at all the prices of all the goods at the same time. Typically, a consumer just looks at the price of a few different goods at a time and relies on his knowledge of approximately how much stuff you can get for a dollar to assess whether prices of the goods he is considering are worth paying. High inflation makes it harder to keep track of how much a dollar is worth, which makes it harder to make the right consumption decisions. It’s a little bit like trying to take measurements with a ruler that keeps changing size. Producers face the same problem: in order to decide what price to charge for the goods they sell, they need to keep track of how much a dollar is worth, and inflation makes this harder.

The government is not the only one to earn seignorage. Part of the money supply is made up of bank deposits, which either pay no interest or pay less than market rates, so they are earning seignorage as well. An additional source of costs of inflation is that it allows banks to earn more seignorage, leading to excessive entry into the banking industry. Exercise 11.7 asks you to compute how much seignorage banks earn.

Exercises

11.1 The Elasticity of Money Demand

Recall the Baumol-Tobin model of Chapter 10. Compute η , the elasticity of money demand with respect to GDP, in the money-demand function that arises from this model.

11.2 The Quantity Theory

The quantity theory of money assumes that money velocity is constant. Try to come up with a money-

demand function that would imply that velocity is indeed constant. What does this money-demand function say about how money holdings depend on interest rates? What does it say about how households manage their trips to the bank?

11.3 Seignorage with Zero Inflation and Growth

Suppose an economy is growing at a constant rate g , money demand comes from the Baumol-Tobin model and the money multiplier is ω . Suppose the government wants to maintain zero inflation. How much seignorage revenue can it obtain as a fraction of GDP? In other words, find an expression for the level of:

$$\frac{M_t^B - M_{t-1}^B}{p_t Y_t}$$

that is consistent with zero inflation. How does this depend on g , F and ω ? Why? [Hint: use the result from Exercise 11.1.]

11.4 Growth in the Money Supply

There are two otherwise identical economies. In both of them GDP is constant, prices are flexible and the real interest rate is constant. The rate of growth of the money supply in each of them is:

	Growth rate of the money supply	
Year	Country A	Country B
2009	3%	6%
2010	3%	5%
2011	3%	4%
2012	3%	3%
2013	3%	2%
2014	3%	1%
2015	3%	0%

How does inflation in the two countries in 2012 compare? Explain.

11.5 Inflation Targeting

- Suppose a Central Bank has decided it wants to keep inflation at exactly 2% every year. The economy is not growing and the real interest rate is 3%. Describe what the Central Bank must do to the money supply to achieve its objective.
- Suppose the real interest rate suddenly and unexpectedly falls from 3% to 1%. How should the Central Bank respond? Explain in detail.

11.6 Seignorage with High Inflation

Suppose that money demand is given by:

$$m^D(Y_t, i_{t+1}) = Y_t \cdot (1 + i_{t+1})^{-\mu}$$

where μ is a parameter. This is sometimes known as the Cagan money-demand function. Equilibrium in the money market is given by:

$$M_t^S = Y_t \cdot (1 + i_{t+1})^{-\mu} p_t$$

The real economy is in a steady state so that:

$$\begin{aligned} Y_t &= Y_{ss} \\ r_{t+1} &= r_{ss} \end{aligned}$$

The government obtains seignorage revenue S_t (in real terms) by expanding the monetary base. S_t is given by:

$$S_t = \frac{M_t^B - M_{t-1}^B}{P_t} \quad (11.4.2)$$

Let ω be the money multiplier, so that $M^S = \omega M^B$.

Suppose that the rate of growth of the money supply is constant at rate γ , i.e.:

$$M_t^S = (1 + \gamma) M_{t-1}^S \quad (11.4.3)$$

- (a) What will be the rate of inflation?
- (b) What will be nominal interest rate?
- (c) Find an expression for the level of real money balances. How does it depend on γ ? Why?
- (d) Solve equation (11.4.3) for M_{t-1}^S and replace this in (11.4.2) to obtain an expression for seignorage revenue S_t in terms of γ and $\frac{M_t^S}{P_t}$. How does S_t depend on γ ? How does it depend on $\frac{M_t^S}{P_t}$? Why?
- (e) Replace the value of $\frac{M_t^S}{P_t}$ that you found in part (c) into the expression for S_t that you found in part (d) to obtain an expression for seignorage revenues S_t in terms of γ , Y_{ss} , r_{ss} , ω and μ .
- (f) Assume the following parameter values:

$$\begin{aligned} Y_{ss} &= 1 \\ \mu &= 4 \\ r_{ss} &= 0.0025 \\ \omega &= 5 \end{aligned}$$

Note that $\mu = 4$ is close to the value that has been empirically estimated in some high inflation countries when the time period is one month. To be consistent, r_{ss} should be interpreted as a monthly real interest rate and Y_{ss} as monthly GDP.

- (g) Plot S_t against γ for values of γ between 0 and 0.8.
- (h) What is the monthly rate of growth of the money supply that maximizes seignorage revenue? Call this γ^* .

- (i) Why is seignorage revenue decreasing in γ for $\gamma > \gamma^*$?
- (j) What monthly inflation rate does $\gamma = \gamma^*$ imply?
- (k) What monthly nominal interest rate does $\gamma = \gamma^*$ imply?
- (l) What yearly rate of inflation does $\gamma = \gamma^*$ imply?
- (m) What is $\frac{M_t}{P_t}$ if $\gamma = \gamma^*$?
- (n) What is the maximum revenue as a fraction of GDP that the government in this example can constantly collect from seignorage?
- (o) Look at the data from Sargent (1982) on the German hyperinflation of the 1920s (you can find it at the book website). What was the maximum monthly inflation rate that Germany experienced? How does it compare to γ^* ?
- (p) Suppose that the government is increasing the money supply at a rate $\gamma = \gamma^*$, and in month t it credibly announces that (i) M_{t+1}^B will not be equal to $M_t^B (1 + \gamma^*)$ the way it would have been if the policy had continued as usual, but it will be some other level $\bar{M}$ (maybe higher, maybe lower than $M_t^B (1 + \gamma^*)$) (ii) from then on, it will set $M_{t+s}^B = \bar{M}$ for every $s \geq 1$ (i.e. the money supply will be constant).
- (q) What will be the rate of inflation from period $t + 1$ onwards?
- (r) What will be the price level P_{t+1} in period $t + 1$? (This will depend on the government's choice of $\bar{M}$)

Suppose the government sets $\bar{M}$ at the level that will ensure that $P_{t+1} = P_t$ (i.e. the level that will immediately stop inflation).

- (s) What value of $\bar{M}$ will it need to choose? How does it compare to $M_t^B (1 + \gamma^*)$?
- (t) How much seignorage revenue will the government obtain in period $t + 1$? How does this level of seignorage compare to what the government was getting every month before making this change?
- (u) Look carefully at the timing of the end of hyperinflation in Germany? When exactly did the money supply stop growing? When exactly did hyperinflation stop? How can we make sense of this?

11.7 Bank Seignorage

Suppose that the demand for M1 money is given by $\frac{M}{pY} = A^{-\eta i}$ where M is the quantity of M1 money, pY is nominal GDP, i is the nominal interest rate and A and η are parameters. Households want to hold a fraction χ of their M1 money in the form of cash and a fraction $1 - \chi$ in the form of checking accounts, which earn no interest. Banks earn seignorage by taking checking deposits and investing in assets that earn the nominal interest rate.

- (a) Find an expression for the ratio of total seignorage earned by banks to GDP. Call this ratio s .
- (b) Compute $\frac{\partial s}{\partial i}$. Why does this number depend on η ?

11.4. The Cost of Inflation

- (c) Look up data on M1 and its components in 2018. What is a reasonable value for χ ? What was the value of $\frac{M}{PY}$? If the average nominal interest was 2%, how much seignorage did banks earn as a fraction of GDP?
- (d) If $\eta = 0.2$, how much seignorage would banks earn as a fraction of GDP if the nominal interest rate went up to 3%?

11.8 Real Interest Rates

Define **realized** real interest rates r_{t+1} as:

$$r_{t+1} \equiv i_{t+1} - \pi_{t+1}$$

where i_{t+1} is the nominal interest rate between period t and period $t+1$ and π_{t+1} is the rate of inflation between period t and period $t+1$. If there is uncertainty about what the rate of inflation is going to be, this implies that there is uncertainty about what realized real interest rates will be. Define the **expected** real interest rate as:

$$r_{t+1}^E \equiv i_{t+1} - \mathbb{E}(\pi_{t+1})$$

where $\mathbb{E}(\pi_{t+1})$ is the expected rate of inflation.

Suppose there are two parties to a nominal lending contract: a borrower and a lender. Who benefits when **realized** real interest rates turn out to be higher than **expected** real interest rates? What monetary policy will each of them support?

11.9 Money among Prisoners of War

Read “The Economic Organisation of a P.O.W. Camp”, which you can find at: <https://www.jstor.org/stable/2550133>. Pick one passage out of the article and explain how it relates to the models of money supply, money demand and inflation from Chapters 10 and 11.